KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 8 · MATH

Sophia's Garden Plot

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Before the Seeds Go In

Sophia is helping design her neighborhood's community garden plot. Different vegetables each need a fraction of the space — but the fractions don't share a denominator. Before you meet her, warm up your fraction skills.

1 Match each word to its meaning. Draw a line, or write the matching letter next to each word.

___ **common denominator**	**A.** A smart, quick guess made by rounding first.
___ **unlike denominators**	**B.** A denominator that two or more fractions can share.
___ **equivalent fraction**	**C.** The bottom number in a fraction.
___ **numerator**	**D.** A fraction that names the same amount, written with different numbers.
___ **denominator**	**E.** Fractions whose bottom numbers are different.
___ **estimate**	**F.** The top number in a fraction.

2 Find a common denominator for 1/2 and 1/3, then rewrite both as equivalent fractions using that denominator. Show your work.

3 Have you ever helped plant a garden, or split up space for different things? Write 1–2 sentences about it.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 8 · MATH

Sophia's Garden Plot

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

After the Garden Is Mapped

Great work on today's lesson! Now show off what you learned about adding and comparing fractions with unlike denominators.

1 Sophia adds a THIRD vegetable: carrots take 1/6 of the plot, on top of tomatoes (1/3) and peppers (1/4). How much of the plot is now used? Show your common denominator work.

2 Which is bigger: 3/8 or 2/5? Show your work using a common denominator.

3 Extension: 5/6 − 1/4 = ? Show every step.

★ **Challenge:** Design your own garden plot! Pick three fractions with different denominators for three plants, add them together, and show all your work.

Looking Ahead: Next week we multiply DECIMALS at Elena's farmers market stand! Bring this sheet back to your next lesson.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 8 · READING

Ravi's Newsletter

Name: _____ Date: _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Before the Newsletter Goes Out

Ravi helps put together his neighborhood's monthly newsletter. Every piece in it is written for a different reason. Before you sort them, warm up your reading brain.

1 Match each word to its meaning. Draw a line, or write the matching letter next to each word.

___ **author's purpose**	**A.** Specific facts or details that support a point.
___ **inform**	**B.** The reason a writer chose to write a piece — usually to inform, persuade, or entertain.
___ **persuade**	**C.** The perspective a writer or speaker sees a topic from.
___ **entertain**	**D.** To convince someone to think or act a certain way.
___ **point of view**	**E.** To give facts or explain something clearly.
___ **evidence**	**F.** To amuse or interest the reader, often through humor or story.

2 Read this sentence: “Recycling saves our planet — start today!” Is this meant to inform, persuade, or entertain? Explain your answer.

3 Think of something you've read — an ad, a joke, a news article. What was its purpose? Write 1–2 sentences.

KCE PATHWAYS · GRADES 4–5 · UNIT 2: THE NEXT LEVEL · WEEK 8 · READING

Ravi's Newsletter

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

After Sorting the Sections

Great work on today's lesson! Now show off what you learned about identifying an author's purpose.

1 List the four newsletter pieces from the lesson and the purpose (inform, persuade, or entertain) of each one.

2 Explain in 2–3 sentences why the bakery spotlight piece blends TWO purposes at once.

3 Why does knowing an author's purpose help you read more carefully? Explain your thinking in 2–3 sentences.

★ **Challenge:** Write THREE short pieces of your own about the same topic — one to inform, one to persuade, and one to entertain.

Looking Ahead: Next week we put TWO different sources together into one bigger idea — that's called synthesizing! Bring this sheet back to your next lesson.

KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 8 · SCIENCE

Earth, Sun and Moon

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. Why do you think we have day and night? Explain your idea.

2. Draw the Sun, Earth and Moon. Use arrows to show how you think each one moves.

3. Predict: Is Earth closer to the Sun in summer or in winter? Why do you think so?

4. Why do you think the Moon seems to change shape during the month?

5. What is one thing you wonder about the Sun or the Moon?

KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 8 · SCIENCE

Earth, Sun and Moon

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use evidence from today's lesson to explain your thinking.

1. Explain the difference between rotation and revolution. What does each one cause?

2. Use two facts from the data table to explain why distance from the Sun does not cause the seasons.

3. Draw the Sun, Earth and the Moon at full moon. Label each one and shade the Moon's dark half.

4. Benjamin Banneker taught himself astronomy. What tools and evidence did he use, and why were his almanacs useful to people?

★ **Challenge:** Keep a Moon journal for one week. Each evening, draw the Moon's shape and write the time. Is it waxing or waning? Predict the shape you will see one week later.

Looking Ahead: Next week we travel out to the planets and stars: gravity, the Sun as a star, and how many cultures read the night sky. Question to bring: If the Sun is a star, why does it look so different from the stars at night?

KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 8 · SOCIAL STUDIES

The American Revolution: Freedom for Whom?

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. What do you already know about the American Revolution? Write two things.

2. Why do you think colonists wanted to break away from Britain? Write your best guess.

3. Prediction: The Declaration says "all men are created equal." In 1776, who do you think was included in that promise? Who might have been left out?

4. Do you think everyone living in the colonies in 1776 wanted independence from Britain? Why or why not?

5. Write one question you have about the American Revolution.

KCE PATHWAYS · GRADES 4-5 · UNIT 2: THE NEXT LEVEL · WEEK 8 · SOCIAL STUDIES

The American Revolution: Freedom for Whom?

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

What Did I Learn?

Use evidence from today's lesson. Write in complete sentences.

1. Name two reasons colonists rebelled against Britain.

2. Using the 1777 petition as evidence, explain how enslaved people used the Declaration's ideas to demand freedom.

3. Choose two people or groups from today (for example, James Armistead Lafayette, Deborah Sampson, Bernardo de Gálvez, the Oneida Nation). Explain which side each chose and why.

4. Freedom for whom? Name one group that gained freedom from the Revolution and one group that did not. Give evidence.

★ **Challenge:** Rewrite "all men are created equal" so that it clearly includes everyone. Then explain in one sentence why the original words still mattered to people who were left out.

Looking Ahead: Next week we study how leaders wrote the U.S. Constitution in 1787 and set up three branches of government. Bring this question: after the Revolution, who should get to make the rules for the new country?